Radial Dimensionality Theory II: White Dwarf Structure under Solar-Constrained Density-Dependent Dimensionality

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We propagate the revised nominal solar parameter sets of Paper I through an audited $n = 3/2$, $\mu_e = 2$ white-dwarf polytropic prescription. The source-precision solar-a score has a sampled amplitude envelope .00874717, locally refined to .00875020; these are conditional model/proxy thresholds, not calibrated observational confidence bounds. We evaluate 66 exponent/ boundary points, 12 central densities and five amplitude fractions within each nominal slice, totaling 3960 models. The largest sampled exact fixed-mass radius deficit is .954933%, at comparison mass $M_A = .848321 M_\odot$, $\alpha = .5$, $A = .00673493$ and $\rho_c = 3 \times 10^6$ g cm^{-3}. This high-density point lies beyond the nonrelativistic EOS applicability estimate and is a mathematical polytrope comparison. The figure uses the actual comparison mass and $\exp(\Delta \ln R) - 1$, rather than a baseline mass label and log shift. The sampled response decreases with amplitude but does not certify a global continuum envelope or universal contraction. Earlier V4 numerical results remain superseded; realistic stellar responses and measurement systematics are needed for quantitative empirical white-dwarf constraints.

VERSION NOTE: V6

This version retains the published title. It propagates the revised solar parameter sets and corrects the fixed-mass comparison and exact percentage shifts. The results describe sampled polytropic white-dwarf models within the stated equation-of-state limits; they do not establish general predictions for realistic white dwarfs.

Scope. $A_{\Delta Q=2.71}(\alpha)$ denotes Paper I's nominal fixed-exponent endpoint, with $\rho_0 = 150$ g cm^{-3} and the solar-a reference prescription. Its sampled/local maxima .00874717/.00875020 have no calibrated confidence coverage. The asymptote associated with the latter is approximately 2.97375; finite-density dimensions require the full law. The same input parameters specify this white-dwarf model without retuning, but do not establish the accuracy of its EOS or modified response. Empirical tests of a specified phenomenological model do not require external endorsement.

I. INTRODUCTION

Paper I of this series [1] formulated Radial Dimensionality Theory (RDT) as a phenomenological law in which the effective spatial dimensionality felt by the gravitational field decreases weakly with local density,

$$d_{\rm eff}(\rho) = 3\,\Omega(\rho), \quad \Omega(\rho) = 1 - A\,\frac{x}{1+x}, \qquad x = \left(\frac{\rho}{\rho_0}\right)^{\alpha}. \tag{1}$$

with $\rho_0 = 150$ g cm^{-3} fixed as a series convention, and derived the corresponding exact modified Lane–Emden equation with the modification confined to the gravitational divergence operator. Confronting solar neutrino fluxes and a helioseismic proxy bound, Paper I found only a small combined-score improvement over the null and obtained a nominal sampled/local amplitude envelope

$$A_{\rm sampled} = .00874717, \quad A_{\rm local} \simeq .00875020, \tag{2}$$

for sampled exponents $\alpha \in [.02, .5]$, without confidence coverage. The associated high-density asymptote is approximately 2.97375.

This paper asks what that bound permits in the next density regime: white dwarfs, with central densities $\rho_c \sim 10^5$–10^6 g cm^{-3}, three to four orders of magnitude beyond the solar core. Because the law of Eq. (1) is a function of density alone, the solar constraint propagates to white dwarfs with *no retuning*: each chosen parameter point specifies a white-dwarf calculation under the adopted polytropic assumptions. We sample each nominal slice at five amplitude fractions, including its endpoint; a global continuum envelope still requires a separate domain check.

This version supersedes an earlier analysis (V4) of white dwarfs in this series. As documented in Sec. IV C, the V4 numerical implementation was invalid, and its modification ansatz differed from the operator form adopted by the series; neither its quantitative predictions (percent-level radius *increases*) nor its Chandrasekhar-mass shifts survive. The retained operator calculation gives radius *deficits* near one percent at fixed mass in the sampled polytropes; the numerical audit below supports those samples, not a global observational bound.

II. FRAMEWORK

A. Modified structure equation

Following Paper I, physical space remains three-dimensional for volume elements, mass integrals, and

microphysics; only the gravitational divergence operator carries $d_{\rm eff}$, with reference scale $r_\star = a$ (the Lane–Emden length) so that the model is unit-invariant. This fixes a physical prescription: changing $r_\star$ at fixed stellar structure would change the variable-δ operator. The length a is spatially constant for each solution but changes with ρ_c across the family. Density here is Newtonian matter mass per ordinary three-dimensional volume. For a polytrope $P = K\rho^{1+1/n}$, $\rho = \rho_c\theta^n$, $\xi = r/a$, the exact structure equation is

$$\theta'' + \left[\frac{2+\delta(\xi)}{\xi} + \frac{d\delta}{d\xi}\ln\xi\right]\theta' + \theta^n = 0, \tag{3}$$

with $\delta = 3[\Omega(\rho_c\theta^n) - 1] \le 0$ and $d\delta/d\xi = (d\delta/d\theta)\,\theta'$, and the regular center condition $\theta \simeq 1 - \xi^2/(2[3+\delta_0])$. Equation (3) is the same equation solved in Paper I; only the polytropic index and the scale relations differ.

B. White dwarf model and validity range

Cold, non-relativistic degenerate matter gives an $n = 3/2$ polytrope with the physical constant

$$P = K\rho^{5/3}, \qquad K = \frac{1.0036\times10^{13}}{\mu_e^{5/3}}\ \text{(cgs)}, \tag{4}$$

for which we take $\mu_e = 2$ (carbon–oxygen composition) [2, 3]. White dwarfs form a one-parameter family in central density: for each ρ_c, $a^2 = (n+1)K\rho_c^{1/n-1}/4\pi G$, $R = a\,\xi_1$, and $M = 4\pi a^3\rho_c\int\xi^2\theta^n d\xi$.

The non-relativistic approximation becomes inadequate as the electron Fermi momentum approaches relativistic values, at $\rho \sim 2\times10^6$ g cm^{-3} for $\mu_e = 2$. The retained numerical grid spans $\rho_c \in [10^5, 3\times10^6]$ g cm^{-3}, with standard-model mass labels $M_0 \simeq 0.16$–$0.86 M_\odot$. Its upper portion therefore extends beyond that estimate and must be read as a mathematical polytrope comparison, not a controlled non-relativistic stellar prediction. There is no solar-style mass or radius recalibration: K, μ_e, and the input central density are held fixed in each pair. Finite-temperature, Coulomb, and envelope corrections are neglected; all results are differential (modified vs. standard model at identical ρ_c and EOS), which does not establish cancellation of these inadequacies in the response. Because the $n = 3/2$ mass–density relation $M \propto \rho_c^{1/2}$ is monotonic with no maximum, this framework makes *no* statement about the Chandrasekhar mass; a meaningful $M_{\rm Ch}$ analysis requires the relativistic equation of state and is outside this calculation.

C. Observable convention

White dwarf tests compare radii at fixed mass. With the standard $n = 3/2$ relation $d\ln R/d\ln M = -1/3$, the fixed-mass radius shift can be written using the fixed-ρ_c shifts as

$$\Delta\ln R\big|_M = \Delta\ln R\big|_{\rho_c} + \tfrac{1}{3}\,\Delta\ln M\big|_{\rho_c}. \tag{5}$$

This identity is exact for comparison at the modified mass $M_A(\rho_c)$: $R_0(M) \propto M^{-1/3}$ implies $\Delta\ln R|_M = \ln[R_A(\rho_c)/R_0(M_A(\rho_c))]$. It is not merely a first-order correction within this polytropic model. The archived figure used $M_0(\rho_c)$ as its horizontal label; the new figure uses $M_A(\rho_c)$. A comparison at that baseline target mass requires solving for a different modified central density. In six audited examples this changes the log radius shift by at most 1.6×10^{-5}. The exact fractional shift is $\exp(\Delta\ln R|_M) - 1$.

III. NUMERICAL IMPLEMENTATION AND VALIDATION

The new propagation uses the flux-form implementation audited against the stated operator in the preceding consistency review. With

$$\begin{gathered} w = \xi^{2+\delta(\theta)}, \quad q = -w\theta', \\ \theta' = -q/w, \quad q' = w\theta^{3/2}, \quad I' = \xi^2\theta^{3/2}, \end{gathered} \tag{6}$$

we integrate ordinary mass as an ODE state. Weighted flux q is not ordinary matter mass for $A \neq 0$. The full density law is retained without suppressing its derivative near the surface. RK45 uses relative/absolute tolerances $2\times10^{-11}/2\times10^{-13}$, maximum step .01 and central start $\xi_0 = 10^{-6}$. Regular seeds are $\theta = 1 - \xi_0^2/[2(3+\delta_c)]$, $q = \xi_0^{3+\delta_c}/(3+\delta_c)$ and $I = \xi_0^3/3$. Solver success, a terminal surface and finite outputs are required.

The null reproduces $\xi_1 = 3.6537537362$ and $I = 2.7140551222$. The earlier audit reproduced all 108 archived paired models using this independently transcribed flux system. It also checked analytic mass/radius scalings, unit changes and central-start convergence. Its strict replay comparison with the old production archive missed a 2×10^{-8} criterion (difference 6.42×10^{-8}), and four tightened derivative- cutoff production solves failed. Those historical issues are retained; the new run does not silently repair or relabel the archived data.

The new inputs are all 65 source-precision solar-a endpoints plus the local solar-envelope peak, with amplitudes at fractions $0, .25, .5, .75, 1$ of each endpoint. Each is evaluated at 12 logarithmic central densities between 10^5 and 3×10^6 g cm^{-3}. All 3960 model rows complete. At 12 selected boundary/density pairs, tightening the flux solver by ten and reducing the maximum step to .003 changes the exact fractional radius shift by at most 2.78×10^{-10}; moving the central start to 10^{-4} changes it by at most 8.05×10^{-11}. No negative-sign requirement is used to accept a model. These comparisons support the sampled effect size; they do not validate the EOS approximation or prove domain extrema.

TABLE I. Selected boundary samples at $\rho_c = 3 \times 10^6$ g cm^{-3}. The last column is an exact percentage shift at M_A, not a log shift. All rows in this table exceed the non-relativistic applicability estimate.

α	$A_{\Delta Q=2.71}$	$\Delta R/R$ (%)
0.02000	0.004311	-0.34306
0.04472	0.005614	-0.48079
0.10000	0.008367	-0.82368
0.22361	0.006711	-0.81111
0.50000	0.006735	-0.95493

IV. RESULTS

A. Sampled conditional signatures

The largest sampled magnitude over the 3960 models is

$$\left.\frac{\Delta R}{R}\right|_{M_A} = -.009549333, \tag{7}$$

at $\alpha = .5$, $A = .006734928$, $\rho_c = 3 \times 10^6$ g cm^{-3} and $M_A = .84832117 M_\odot$. The standard mass at that input density is $.85925632 M_\odot$, illustrating why the mass-coordinate correction matters. The entire boundary set spans comparison masses $.15502$–$.85539 M_\odot$. The largest-deficit point is a mathematical high-density polytrope example, not a controlled non-relativistic prediction for a real white dwarf. At every sampled exponent and density the radius shift decreases over the five amplitude fractions; the largest sampled magnitude occurs at a boundary point. This is a finite-grid observation, not a proof of monotonicity between samples or over all accepted parameters. The zero-amplitude models have zero shift and arbitrarily small effects remain permitted. The new numbers come from the revised solar region, rather than rescaling the older .95% example.

The curves vary moderately with mass under this polytropic prescription. That behavior does not follow from saturation over the full exponent range: small-exponent models remain appreciably unsaturated at white-dwarf densities. The full density law and its operator contribution are retained.

B. Sign and physical interpretation

The reported sampled models have radius deficits at fixed mass. The term $(2 + \delta)/\xi$ alone does not prove a universal sign because $\delta' \ln \xi$ and calibration also contribute. Paper I holds the physical solar radius fixed and shows both signs of the shift in ξ_1; it does not establish a physical solar contraction. V4's force multiplier $F = (d_{\text{eff}} - 1)/2$ is a different prescription. A sign change between prescriptions is not by itself a validation criterion.

C. Why the V4 results were invalid

For the record, the V4 implementation failed independently of the ansatz choice: (i) it set the polytropic constant $K = 1$ in cgs units, making its model stars $\sim$10 m objects (R = 430–926 cm over the scanned densities); (ii) it began integration at $r_0 = 10^3$ cm—*outside* those stars—seeding the integration with a uniform-density sphere that dominated the result (computed masses 9–79$\times$ the analytic values, and mass–radius scalings inconsistent with $R \propto M^{-1/3}$); (iii) its "Chandrasekhar mass shifts" were maxima over a scan grid of a monotonic $M(\rho_c)$ relation—the $n = 3/2$ polytrope has no maximum mass—and so measured the grid edge, not physics. The validation gates of this version were designed directly against these failure modes.

V. DISCUSSION

A. Observability

The mathematical sampled scale of roughly one percent can be compared with the precision of white dwarf mass–radius measurements. Eclipsing double-lined binaries deliver model-independent masses and radii at the 1–2% level [4], and *Gaia* parallaxes constrain the photometric mass–radius relation for large samples [5, 6]. Two practical points temper the test. First, the sampled deficit varies moderately with mass and can be degenerate with global systematics of the mass–radius relation (composition, envelope thickness, finite temperature) rather than appearing as a distinctive shape; distinguishing it requires either absolute accuracy at the few$\times 10^{-3}$ level or a composition- and temperature-controlled differential sample. Second, these reported sample scales are not lower bounds: the model permits arbitrarily small shifts. A quantitative exclusion or detection forecast requires composition, thermal and envelope uncertainties, correlated measurement errors, and checked modified stellar responses within the specified EOS and structure assumptions. A null sample does not automatically provide a stronger bound than the solar proxy.

B. Relation to the series

The Sun supplies a conditional model/proxy threshold. White dwarfs and neutron stars probe different structural and relativistic assumptions; sharing a density law alone neither validates their responses nor removes EOS degeneracy. Saturation and sign must be checked over the actual domain.

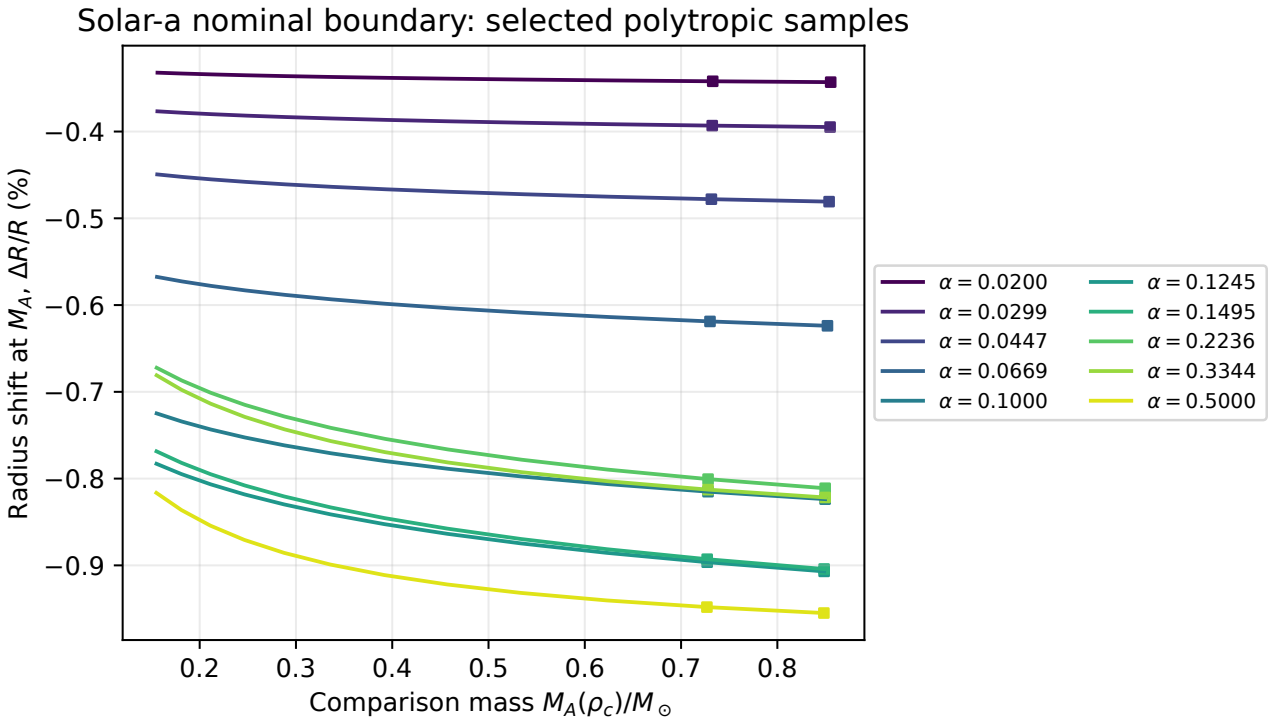


FIG. 1. Selected curves from the 66 nominal solar-a boundary points. The abscissa is the actual modified comparison mass $M_A(\rho_c)$; the ordinate is the exact percentage shift $100[\exp(\Delta \ln R|_{M_A}) - 1]$. Square markers identify $\rho_c > 2 \times 10^6$ g cm^{-3}, beyond the stated non-relativistic EOS estimate. The curves are polytropic samples, not certified continuous-domain bounds or measured white-dwarf deficits.

VI. CONCLUSIONS

The revised nominal solar-a sets produce sampled fixed-mass radius deficits up to .954933% in the $n = 3/2$ model. The new figure uses M_A and exact fractional shifts. Sampling interior amplitude fractions supports a monotonic trend on the tested grid, but does not certify a global continuum envelope. The largest-deficit point exceeds the non-relativistic applicability estimate; realistic EOS, thermal, composition and envelope responses remain separate tasks.

The flux-form numerical checks support the size of these conditional model responses. V4's quantitative results remain superseded, and this polytrope has no Chandrasekhar maximum. No detection, guaranteed observational sensitivity or calibrated white-dwarf confidence bound is inferred from the solar proxy.

CODE AND REVISION AVAILABILITY

Producing scripts, inputs, saved results and numerical audit evidence are publicly available in the White Dwarfs code repository. The accompanying source package, `whitedwarfs_code`, includes the manuscript, dependencies and resource-controlled reproduction commands. Its `reproduction/CAPSULE.json` records file hashes; `reproduce.py` verifies the package and rebuilds the paper from retained scientific outputs. Full scientific runs are separate from this saved-result reproduction. Historical scripts and results remain separately identified.

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